

Least-squares problem $\min_x \|Ax - b\|$ $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$

If $A = U \Sigma V^T$ is a thin SVD of A , we can write the solution of min. norm as $x = V \Sigma^+ U^T b$, $\Sigma^+ = \begin{bmatrix} 1/\sigma_1 & 1/\sigma_2 & \dots & 1/\sigma_r & 0 & \dots \end{bmatrix}$, where $r = \text{rank}(A)$.

$$A = \sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T + \dots + \sigma_r u_r v_r^T$$

$$U = \begin{bmatrix} u_1 & u_2 & \dots & u_r \end{bmatrix} \quad V = \begin{bmatrix} v_1 & v_2 & \dots & v_r \end{bmatrix}$$

$$x = \left(\frac{1}{\sigma_1} v_1 u_1^T + \frac{1}{\sigma_2} v_2 u_2^T + \dots + \frac{1}{\sigma_r} v_r u_r^T \right) b$$

Rmk: The small singular values of A influence a lot the solution of the least-squares problem. If we want to avoid this, we can "truncate" the singular values which are smaller than a certain tolerance (say, 10^{-5}).

Tikhonov regularization: $x_{\text{Tik}} := \argmin_{x \in \mathbb{R}^n} \|Ax - b\|^2 + \alpha \|x\|^2$ for some fixed (small) parameter $\alpha \in \mathbb{R}$.

$$= \argmin_{x \in \mathbb{R}^n} \left\| \begin{bmatrix} A \\ \sqrt{\alpha} I \end{bmatrix} x - \begin{bmatrix} b \\ 0 \end{bmatrix} \right\|^2 = \argmin_{x \in \mathbb{R}^n} \left\| \begin{bmatrix} Ax - b \\ \sqrt{\alpha} x \end{bmatrix} \right\|^2$$

$$= \argmin_{x \in \mathbb{R}^n} \|Ax - b\|^2 + \alpha \|x\|^2$$

Note: $\tilde{A} \in \mathbb{R}^{(n+m) \times n}$ has full column rank (even if A doesn't!)

$$\Rightarrow x = (\tilde{A}^T \tilde{A})^{-1} \tilde{A}^T \tilde{b} = \left(\begin{bmatrix} A^T & \sqrt{\alpha} I \end{bmatrix} \begin{bmatrix} A \\ \sqrt{\alpha} I \end{bmatrix} \right)^{-1} \begin{bmatrix} A^T & \sqrt{\alpha} I \end{bmatrix} \begin{bmatrix} b \\ 0 \end{bmatrix}$$

$$= (A^T A + \alpha I)^{-1} A^T b$$

Let's see how we can write it in terms of the thin SVD of $A = U \Sigma V^T$

$$A^T A + \alpha I = V \underbrace{\Sigma U^T U \Sigma^T}_{=I \text{ because } U \text{ has orthonormal columns}} + \alpha I = V \Sigma^2 V^T + \alpha I = V \Sigma^2 V^T + \alpha V V^T$$

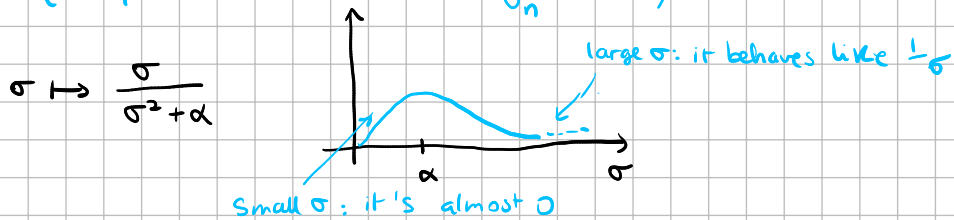
$$= V (\Sigma^2 + \alpha I) V^T = V \cdot \begin{bmatrix} \sigma_1^2 + \alpha & & \\ & \sigma_2^2 + \alpha & \\ & & \ddots \\ & & & \sigma_n^2 + \alpha \end{bmatrix} V^T$$

$$(A^T A + \alpha I)^{-1} = V \cdot \begin{bmatrix} \frac{1}{\sigma_1^2 + \alpha} & & \\ & \ddots & \\ & & \frac{1}{\sigma_n^2 + \alpha} \end{bmatrix} V^T$$

$$x = (A^T A + \alpha I)^{-1} A^T b = V \begin{bmatrix} \frac{1}{\sigma_1^2 + \alpha} & & \\ & \ddots & \\ & & \frac{1}{\sigma_n^2 + \alpha} \end{bmatrix} V^T V \Sigma U^T b = V \begin{bmatrix} \frac{\sigma_1}{\sigma_1^2 + \alpha} & & \\ & \ddots & \\ & & \frac{\sigma_n}{\sigma_n^2 + \alpha} \end{bmatrix} U^T b$$

$$= \left(\frac{\sigma_1}{\sigma_1^2 + \alpha} v_1 u_1^T + \dots + \frac{\sigma_n}{\sigma_n^2 + \alpha} v_n u_n^T \right) b$$

Before: $\left(\frac{1}{\sigma_1} v_1 u_1^T + \dots + \frac{1}{\sigma_n} v_n u_n^T \right) b$



comment: it's a "smooth" version of the truncated SVD discussed before.

CONDITIONING, SENSITIVITY, ACCURACY OF ALGORITHMS.

[Matlab example: $A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 4 \\ 1 & 2 & 3+\epsilon \end{bmatrix}$ $\leadsto$ QR and SVD ok, normal eq. definitely NOT ok!]

Condition number of a problem: measures how much the output can change when you perturb the input.

Def: If $F: \mathbb{R}^n \rightarrow \mathbb{R}^n$ the "ABSOLUTE CONDITION NUMBER" is the smallest number $K_{abs}(F, x)$ such that

$$\|F(x+\delta) - F(x)\| \leq \|\delta\| \cdot K + O(\|\delta\|^2)$$

Example: $F: \mathbb{R} \rightarrow \mathbb{R}$ $F(x) = x^2$
 $F(x+\delta) - F(x) = (x+\delta)^2 - x^2 = 2x\delta + \delta^2$
 $|F(x+\delta) - F(x)| \leq \underbrace{|2x| \cdot |\delta|}_{K_{abs}(F, x)} + O(\delta^2)$

$F: \mathbb{R} \rightarrow \mathbb{R}$ $F(x+\delta) = F(x) + \delta F'(x) + O(\delta^2)$
 $\leadsto$ for $F \in C^2(\mathbb{R})$, $K_{abs} = |F'(x)|$
 for $F: \mathbb{R}^n \rightarrow \mathbb{R}^n$, $K_{abs}(F, x) = \|\nabla F(x)\|$

Rmk: relative errors are (most of the times) much more meaningful than absolute errors.

$$\|x - \tilde{x}\| \quad \text{vs.} \quad \frac{\|x - \tilde{x}\|}{\|x\|}$$

$$\tilde{x} = x + \delta \quad y = F(x) \quad \tilde{y} = F(\tilde{x})$$

$$\|\tilde{y} - y\| \leq K_{abs}(F, x) \cdot \|\tilde{x} - x\| + O(\|\tilde{x} - x\|^2)$$

$$\frac{\|\tilde{y} - y\|}{\|y\|} \leq \underbrace{\frac{K_{abs}(F, x) \|x\|}{\|y\|}}_{K_{rel}(F, x)} \cdot \frac{\|\tilde{x} - x\|}{\|x\|} + \text{smaller order terms}$$

" $K_{rel}(F, x)$ RELATIVE CONDITION NUMBER"

high condition number $\leadsto$ "ill conditioned" $\leadsto$ difficult problem

Small condition number $\leadsto$ "well conditioned" $\leadsto$ easier problem

Linear systems: $Ax = b$ $A \in \mathbb{R}^{n \times n}$ $b \in \mathbb{R}^n$

1) condition number of the problem with respect to perturbations of b

Instead of $Ax = b$, we solve $A\tilde{x} = \tilde{b}$
 $\downarrow$ $\downarrow$
 $x = A^{-1}b$ $\tilde{x} = A^{-1}\tilde{b}$

$$\|\tilde{x} - x\| = \|A^{-1}\tilde{b} - A^{-1}b\| = \|A^{-1}(\tilde{b} - b)\| \leq \|A^{-1}\| \cdot \|\tilde{b} - b\|$$

$$\|b\| = \|Ax\| \leq \|A\| \cdot \|x\| \rightsquigarrow \|x\| \geq \frac{\|b\|}{\|A\|} \rightsquigarrow \frac{1}{\|x\|} \leq \frac{\|A\|}{\|b\|}$$

$$\Rightarrow \frac{\|\tilde{x} - x\|}{\|x\|} \leq \underbrace{\|A\| \cdot \|A^{-1}\|}_{\sigma_1(A)/\sigma_n(A)} \cdot \frac{\|\tilde{b} - b\|}{\|b\|}$$

$$\sigma_1(A)/\sigma_n(A)$$

but it's what we already defined a long time ago as the condition number of the matrix A !

2) condition number of the problem with respect to perturbations of A

$$\tilde{A} = A + E \quad Ax = b \rightarrow x = A^{-1}b \quad (A+E)\tilde{x} = b \rightarrow \tilde{x} = (A+E)^{-1}b$$

Let δ be s.t. $\tilde{x} = x + \delta$

$$b = (A+E)\tilde{x} = (A+E)(x+\delta) = \underbrace{Ax}_b + A\delta + Ex + E\delta$$

$$\rightsquigarrow A\delta + Ex + E\delta = 0$$

$\downarrow$ (small thing) · (small thing) $\rightarrow$ very small and can be ignored

$$\delta \approx -A^{-1}Ex$$

$$\begin{aligned} \frac{\|\tilde{x} - x\|}{\|x\|} &= \frac{\|\delta\|}{\|x\|} \approx \frac{\|A^{-1}Ex\|}{\|x\|} \leq \frac{\|A^{-1}\| \cdot \|E\| \cdot \|x\|}{\|x\|} \\ &= \|A^{-1}\| \cdot \|\tilde{A} - A\| \cdot \frac{\|A\|}{\|A\|} = (\underbrace{\|A^{-1}\| \cdot \|A\|}_{\text{we got } \kappa(A) \text{ again!}}) \cdot \frac{\|\tilde{A} - A\|}{\|A\|} \end{aligned}$$

[singular matrix = not invertible]

Theorem: $\frac{1}{\kappa(A)} = \min_{\substack{X \in \mathbb{R}^{n \times n} \\ \text{Singular}}} \frac{\|X - A\|}{\|A\|}$

$\|X - A\|$ is the distance between A and X

$\min_{X \text{ singular}} \|X - A\|$ is the distance of A from the set of non-invertible matrices

$\min_{X \text{ singular}} \frac{\|X - A\|}{\|A\|}$ is rel. dist "

if $\kappa(A)$ is very large $\Rightarrow \frac{1}{\kappa(A)}$ very small $\Rightarrow A$ is very close to being not invertible

Proof: $\min_{X \text{ singular}} \|X - A\| = \min_{\text{rank}(X) \leq n-1} \|X - A\|$

$= \sigma_n(A).$

Eckart-Young

theorem: a solution is obtained by throwing away the smallest sing. value of A

$$\Rightarrow \frac{\min_{X \text{ singular}} \|X - A\|}{\|A\|} = \frac{\sigma_n(A)}{\sigma_1(A)} = \frac{1}{\kappa(A)}.$$

□

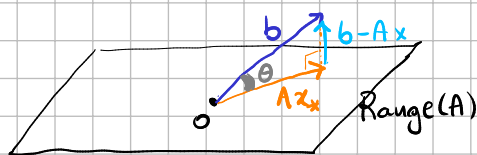
what about least-squares problems?

Theorem: $A \in \mathbb{R}^{m \times n}$ with full column rank, $\min_x \|Ax - b\| \leadsto$ with solution x_* .

Let $\cos \theta = \frac{\|Ax_*\|}{\|b\|}$, then the relative cond. number of the problem

wrt b is $\kappa_{\text{rel}, b \rightarrow x} \leq \frac{\kappa(A)}{\cos \theta}$ and the rel. cond. num. of

the problem wrt A is $\kappa_{\text{rel}, A \rightarrow x} \leq \kappa(A) + \kappa(A)^2 \cdot \tan \theta$



• if θ is large, ^{close to 90°} small changes in b result in potentially large changes in the solution

• if $\theta \approx 0$ then $\kappa_{\text{rel}, A \rightarrow x} \lesssim \kappa(A)$ (the same as linear systems)

• otherwise, it depends on $\kappa(A)^2 \dots$